Autonomous Naval Epoxy-Polyamide Formulation Engine

Customer: a top U.S. Navy prime Decision packet Recommended qualification path: Lineage-60 benchmark + CRO falsification

Authority lock: Candidate only: not QPL-listed, not NAVSEA-qualified, and not an approval for procurement. Qualification status must be verified against QPL-24441; the DLA page retrieved 2026-07-03 reports last update 27-JAN-2026.

One-line status: The engine generated a cited, arithmetic-feasible, CRO-screen-ready epoxy-polyamide candidate package pending supplier declarations. Arithmetic recomputes; pot-life and corrosion are bench-gated and not predicted; zero wet-lab data exists; sourcing is blocked on PFAS/TSCA/TAA/SDS/TDS/CoA declarations; no QPL/NAVSEA qualification is claimed.

1. KPI dashboard

KPI	Expected model behavior	Current result	Status	Conf	Next falsifier
Envelope arithmetic	hit 1:1 vol, 65% VS, VOC≤340, WPG, PVC, coverage	VS 65%, VOC 282 g/L, WPG 9.16; audit 6/6	PASS	High	Method 24 / D2697 / D1475
Candidate generation	generate CRO-screen-ready concepts from real materials	4320→2444 survivors; 10 concepts in feasibility bands	PASS	Med	CRO compatibility / KU screen
Pot life	predict/screen ≥1.5 h usable pot life	model FAILED: best 44% within ×/1.5 (bar 80%)	FAIL	Low	week-1 KU-vs-time 65/73/95F
Corrosion (immersion)	test immersion barrier hypothesis	corrosion model has NO AUTHORITY (2 strict in-class series; bar ≥3); barrier directional-only	FAIL	Low	B117/D5894/D870/ EIS + locked falsification
Sprayability	airless / brush / roller	low-shear KU proxy only; high-shear atomization unvalidated	GATE	Low	D562 / D4400 / spray-out
Shelf life	36-month unopened	extrapolated vs qualified analog; no data	GATE	Low	D1849 / D869 accelerated
Sourcing / regulatory	supplier-declaration-gated package	PFAS / TSCA / TAA declarations UNKNOWN (RFQ required, all lines)	BLOCKED	Low	supplier declarations / RFQ
Source quality	cited, value-verified constants	48 single-source; 0 load-bearing violations	LIMITED	Med	value-vs-value reconcile
Qualification	support Navy procurement	NOT claimed: not QPL / NAVSEA-qualified	NOT-CLAIMED	–	authority verification

2. Model rights matrix

Model	Purpose	Right to claim	Next promotion test
M2 VOC/solids/density/mix-r ...	Solve two-component recipes onto 1:1 ...	ARITHMETIC USABLE	Audit by Method 24, D2697, D1475, drawdowns, ...
M12 Resin/curative property ...	Predict resin/curative equivalent ...	EXACT STRUCTURE ONLY	Train ML estimated-tier Tg/viscosity models ...
M5 Known-material ...	Show the engine can rediscover known ...	SANITY CHECK	Run Lineage-60 and a commercial reference ...
M6 Solvency model	Score solvent blend compatibility ...	LIMITED SANITY	CRO compatibility, blush, phase-split, and ...
M7 Viscosity KU proxy	Band sprayability risk from solids ...	LIMITED SANITY	D562/D4400 spray/sag map and yellow D2805 ...
M8 Cure/pot-life model	Predict usable pot-life and cure ...	NO AUTHORITY	Week-1 KU-vs-time and usable-pot-life screen ...
M9 Corrosion rank model	Rank corrosion resistance from public ...	NO AUTHORITY	Side-by-side B117/D5894/D870/ISO 16773 EIS ...
M10 Barrier tortuosity model	Encode barrier design rules and lock ...	DIRECTIONAL ONLY	Register locked rank before panels; compare ...
M15 Bayesian CRO optimizer	Choose next experiments from measured ...	ROADMAP	Run first CRO batch, ingest raw outcomes, ...

Autonomy boundary: **What "autonomous" means here:** human-defined problem envelope; machine-executed source retrieval, constraint solving, gating, scoring, and CRO-test design; human-controlled lab / EHS / legal / procurement. It is deterministic machine-executed search + audit within a human-set envelope — not autonomous science.

3. Track comparison and feasibility bands

Candidate Feasibility Band summary: bands are display groupings only; bench decides pot-life, corrosion, and spray ordering.

Track	Use	VS	VOC	WPG	PVC/Λ	R	ZnP	Status
Lineage-60	Public F160/F161 lineage reconstruction; authority benchmark, not an envelope-literal candidate	60.2%	285	9.24	8.2 / 0.15	0.78	no	major variance: public black/yellow solids are 60.2+/-2%, not 65+/-2%
Envelope-65	Envelope-literal 65% candidate; primary response to the customer envelope	65.0%	282	9.16	9.3 / 0.17	0.70	no	absent from the primary; color pigments are identification only
Enhanced-65	Optional process/corrosion-risk variant; functional zinc phosphate must earn its place at the bench	65.0%	282	9.35	10.9 / 0.20	0.70	yes, functional	functional optional inhibitor; not part of the primary non-functional-pigment response

Feasibility band	Concepts	Score	Meaning
Band A	C01, C02, C03, C04	0.805	score 0.805; C01-C04; indistinguishable by current model; bench decides; no pot-life/corrosion/spray rank authority
Band B	C05, C06, C09, C10	0.803	score 0.803; C05/C06/C09/C10; minor sourcing/score difference only; bench decides; no pot-life/corrosion/spray rank authority
Band C	C07, C08	0.795	score 0.795; C07/C08; lower sourcing robustness; bench decides; no pot-life/corrosion/spray rank authority

These are feasibility bands, not a performance rank. Pot-life, corrosion, and sprayability have no validated rank authority — the bench decides ordering.

4. Envelope-65 recipe summary

Detailed tolerances, scale factors, CoA fields, order of addition, QC targets, stop rules, and returned fields are in envelope65_cro_recipe_card.json.

Variant	Part	Role	g basis	Lines
black	part_A	epoxy side	137.4	dgeba_828, silica, dispersant, defoamer, thixotrope, iron_oxide_black, naphtha
black	part_B	polyamide-curative side	102.2	ancamide_350a, silica, dispersant, thixotrope, n_butanol
yellow	part_A	epoxy side	136.8	dgeba_828, silica, dispersant, defoamer, thixotrope, iron_oxide_yellow, naphtha
yellow	part_B	polyamide-curative side	102.2	ancamide_350a, silica, dispersant, thixotrope, n_butanol

5. First falsification test: does the engine beat simple heuristics?

Pre-registered locked test claim: This can lose. The barrier index locks the order **Lineage-60 > Enhanced-65 > CRO-Robust-65 > Envelope-65** and the decisive pairwise bet is **the Enhanced-65 > Envelope-65 non-obvious call must hold in the measured data (the single most decisive, PVC-contrarian bet).**. Decision rule uses Kendall tau plus the key pairwise bet; third-party registration is required before panels; locked hash 197cf16aa46bca21.... Uncertainty status: no calibrated uncertainty available. Enhanced-65 minus Envelope-65 barrier-index delta is 0.015; deltas are too close to claim performance separation.

Locked order	Track	Barrier index	PVC %	ZnP
1	Lineage-60	0.477	8.2	no
2	Enhanced-65	0.474	10.9	yes
3	CRO-Robust-65	0.462	9.5	no
4	Envelope-65	0.460	9.3	no

Decision handling: Controls fail → repeat/review, not pass. Envelope-65 beats Enhanced-65 → demote the zinc-phosphate hypothesis. Noisy or tied decisive pair → no model promotion. Lineage-60 wins while all 65% tracks fail → use Lineage-60 path; do not qualify 65%.

6. CRO kill gates and returned data

Gate	Method	Target	Fail / inconclusive rule	Raw return field
gel_or_phase_split	visual check plus KU trend during induction and first ...	0 observed gel/phase-split events	repeat/review if control invalid; otherwise stop or reformulate on ...	visual_observation, ku_vs_time_3temps.raw_value, ...
pot_life_below_90_min	ASTM D562 KU-vs-time at 65/73/95F plus application ...	≥90 min at 73F	repeat/review on control failure or noisy tie; stop/reformulate if ...	pot_life_h.raw_value, ku_vs_time_3temps.raw_value, ...
voc_over_340_g_l	EPA Method 24 / ASTM D3960	≤340 g/L	inconclusive if worksheet missing; stop/reformulate if valid value ...	voc_g_l.raw_value, method24_worksheet_path
volume_solids_outside_63_67	ASTM D2697 or lab-accepted Method 24 solids worksheet	63-67%	inconclusive if raw replicates/worksheet missing; stop/reformulate if ...	volume_solids_pct.raw_value, replicate_values, ...
severe_sag_or_pinhole	drawdown + airless spray observation, ASTM D4400/D1210	Hegman ≥6; sag ≥12 mil target	inconclusive if setup missing; stop/reformulate if valid application ...	fineness_hegman.raw_value, sag_mils.raw_value, ...
failed_adhesion	ASTM D4541 dry and post-immersion	≥20% margin vs threshold/control basis where ...	repeat/review on adhesive failure or control failure; stop on valid ...	adhesion_mpa.raw_value, failure_mode, dolly_map_path
early_corrosion_failure	ASTM B117/D1654, ASTM D870, ISO 16773 EIS	no tied/noisy decisive pair; EIS gap target ...	controls fail -> repeat/review, not pass; tied/noisy decisive pair -> ...	eis_immersion.raw_spectra_path, ...

7. Sourcing, regulatory, and source quality

Run manifest: Whole-run output hash e15c20df360d12cd18e42888…; no manual candidate edits after run start attested.

Shelf life remains an extrapolation until ASTM D1849/D869 data exists. Whole-run output hash: e15c20df360d12cd18e42888….

Blocker	Status
RFQ lines	open
PFAS	unknown / RFQ required
TSCA	unknown / RFQ required
TAA/domestic	unknown / RFQ required
Current SDS/TDS/CoA	unknown / RFQ required
Lead time/MOQ/shelf/storage	unknown / RFQ required

KPI	Current value
Constants total	83
Two-source constants	11
Arithmetic-crosschecked constants	8
Single-source constants	48
Manual-required value comparisons	49
Load-bearing violations	0
Confidence	arithmetic HIGH; source values MEDIUM; sourcing LOW

8. Clean-slate provenance summary

Item	User	Engine	Gen	Validation	Guard
Human technical envelope / problem constraints	Y	N	Y	N	N
F160/F161 public MIL formula lineage	N	Y	N	Y	Y
F150/NCP Type IV partial-TDS anchors	N	Y	N	Y	N
Evonik SPF A702B75P1 starting-point formulation	N	Y	N	Y	N
Supplier raw-material families	N	Y	Y	N	N
Corrosion literature	N	Y	N	N	N
QPL/F-718 authority records	N	Y	N	N	Y

Pointers

Known-control checks, full model ledger, corrosion theory, patents, source tables, and candidate card grids are in TECHNICAL_APPENDIX.pdf. Molecule-engine appendix: MOLECULE_ENGINE.pdf. Machine-readable artifacts: clean_slate_provenance.json, envelope65_cro_recipe_card.json, falsification_protocol.json, run_manifest.json.